

Photosynthesis and Plant Biology Quiz

Topic: Photosynthesis and plant biology | Difficulty: Easy | Questions: 5 | Model: gemini-2.5-flash

1. What three main things do plants need to perform photosynthesis?

- A. Sunlight, water, and carbon dioxide (correct)**
- B. Oxygen, soil, and darkness
- C. Sugar, light, and nitrogen
- D. Wind, rain, and sand

Explanation: Photosynthesis is the process where plants use sunlight, water, and carbon dioxide to create their own food.

2. What two important things do plants produce during photosynthesis?

- A. Oxygen and sugar (glucose) (correct)**
- B. Carbon dioxide and water
- C. Nitrogen and sunlight
- D. Soil and minerals

Explanation: Plants convert light energy into chemical energy in the form of sugar (glucose) and release oxygen as a byproduct.

3. In which part of a plant does most photosynthesis take place?

- A. Roots
- B. Flowers
- C. Leaves (correct)**
- D. Stems

Explanation: Leaves contain chloroplasts, which are the organelles where photosynthesis occurs, making them the primary site for this process.

4. What is the name of the green pigment found in plants that captures sunlight for photosynthesis?

- A. Melanin
- B. Hemoglobin
- C. Chlorophyll (correct)**
- D. Carotene

Explanation: Chlorophyll is the green pigment essential for absorbing light energy during photosynthesis.

5. What is the main job of the roots of a plant?

- A. To make food for the plant
- B. To attract pollinators
- C. To absorb water and nutrients from the soil (correct)**
- D. To support the flowers

Explanation: Roots anchor the plant and are crucial for absorbing water and dissolved minerals from the soil.

Answer Key

1. **A** Sunlight, water, and carbon dioxide
2. **A** Oxygen and sugar (glucose)
3. **C** Leaves
4. **C** Chlorophyll
5. **C** To absorb water and nutrients from the soil